

Module 1: Linux Server - Understand and Use Essential Tools

Section 1: Concepts & Definitions

1. What is the minimum number of partitions you need to install Linux?

The absolute minimum is **two**:

- **/** (Root partition): Where the operating system and files reside.
- **Swap**: Used as virtual memory when RAM is full.
(Note: Modern systems can technically run on just one root partition using a "swap file," but for standard installation questions, the answer is usually two.)

2. Explain the Chmod Command.

chmod (change mode) is used to change the access permissions of files and directories. It manages three types of users (**Owner, Group, Others**) and three types of permissions (**Read (4), Write (2), Execute (1)**).

- Example: `chmod 755 file.txt` gives the owner full access and others read/execute access.

7. Describe the root account.

The **root** account is the superuser in Linux. It has the highest level of administrative privileges and can access, modify, or delete any file on the system, regardless of ownership. It has the User ID (UID) of **0**.

8. What is a shell?

A shell is a computer program that acts as an interface between the user and the operating system kernel. It interprets text-based commands entered by the user and tells the OS to execute them.

9. What is Linux?

Linux is an open-source, Unix-like operating system kernel created by Linus Torvalds. When combined with various tools, libraries, and desktop environments, it forms a "Linux Distribution" (like Ubuntu, CentOS, or Kali).

10. What is Bash?

Bash (Bourne Again SHell) is the default shell for most Linux distributions. It is an improved version of the original Sh (Bourne shell) and includes features like command history and tab

completion.

Section 2: Practical Steps & Scenarios

3. How to check Linux memory utilization?

You can use the following commands:

- `free -m` (Shows free and used memory in Megabytes).
- `top` or `htop` (Real-time dynamic view of system resources).
- `vmstat` (Report virtual memory statistics).

5. Get connecting on a Linux server by SSH.

To connect to a remote server, use the command:

`ssh username@server_ip_address`

- *Example:* `ssh root@192.168.1.10`

6. Create 5 files in the /tmp directory, and then use tar and gzip to bundle and compress the files.

1. **Create files:** `touch /tmp/file{1..5}.txt`
2. **Bundle and Compress:** `tar -cvzf backup.tar.gz /tmp/file1.txt /tmp/file2.txt /tmp/file3.txt /tmp/file4.txt /tmp/file5.txt`
(Alternatively: `tar -cvzf backup.tar.gz /tmp/file*.txt`)

11. You have a new empty hard drive that you will use for Linux. What is the first step you use?

The first step is **Partitioning** the drive (using tools like `fdisk`, `gdisk`, or `parted`) to define the space, followed by **Formatting** the partitions with a filesystem (like `ext4` or `xfs`).

Section 3: Essential Linux Commands

Ques #	Goal	Command
4	Search for patterns in files	grep "pattern" filename
12	Show current working directory	pwd
13	Get help with various options	man [command] or [command] --help
14	Display what users are doing	w (or who)
15	Get info about the OS	uname -a or cat /etc/os-release
16	Create a hard link	ln source_file link_name
17	Create a soft link (File/Dir)	ln -s source_target link_name
18	Search for specific pattern	grep "pattern" filename
19	Basic Regex with grep	grep -E "^ [0-9]" file (Searches for lines starting with a number)

